

S-FMM: A nonlinear spatial-frequency modulation framework for structured oscillatory fields on the torus

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ABSTRACT

Understanding and modeling two-dimensional fields is a fundamental challenge in many areas of science, where spatial organization, directionality, and periodicity play a central role. To address this problem, we introduce the Spatial-Frequency Modulated Möbius model (S-FMM), a structured parametric framework for modeling fields defined on the torus. The framework is motivated by the analysis of spatial patterns arising in magnetically confined plasmas, which naturally exhibit periodic geometry and coherent directional modes.

S-FMM represents the field as a compact collection of nonlinear modes defined along two angular coordinates. Möbius-transformed phase functions are used to model non-sinusoidal and anisotropic spatial patterns, allowing the framework to adapt to localized and direction-dependent structures that are difficult to capture with classical spectral methods. Model parameters are estimated through a specifically designed algorithm for signal-plus-noise settings, with identifiability ensured via simple ordering constraints.

The effectiveness of the proposed approach is demonstrated through numerical experiments covering a range of noise conditions and spatial configurations. In particular, an application to high-resolution plasma turbulence simulations obtained from established physics-based models shows that S-FMM successfully recovers coherent directional structures that are poorly represented by classical decomposition methods. Across all tested scenarios, S-FMM achieves higher reconstruction accuracy with fewer parameters, providing a compact and parsimonious representation of the underlying fields that facilitates analysis and dimension reduction of fields defined on periodic domains. The scope of application is not limited to magnetically confined plasmas and naturally includes other settings where toroidal or angular organization is present. For example, S-FMM is well suited for the analysis of grid cell firing patterns in neuroscience, where latent toroidal structure plays a central role.

1. Introduction

Oscillatory patterns arise in a broad spectrum of two-dimensional systems, including physical fields, wave phenomena, periodic structures in engineered surfaces, and biological processes. In many such contexts, the observed data are noisy, sparse, or corrupted, conditions that further complicate an already challenging problem: the reliable identification of multidimensional, spatially structured oscillations. These patterns are inherently hard to detect, often involving complex interactions of direction, frequency, and phase, arising from the superposition of several oscillatory components, that are not easily captured by classical tools. Capturing such behaviors requires models that are not only expressive, but also flexible, parsimonious, and structurally meaningful.

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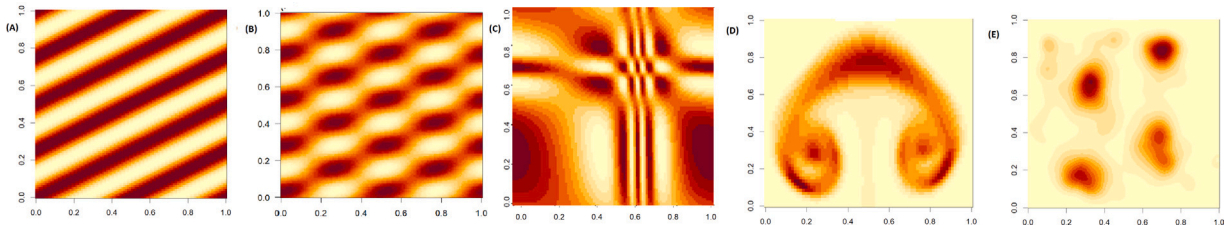


Fig. 1. Examples of two-dimensional oscillatory patterns over toroidal domains illustrating increasing structural complexity: (A) Pure Fourier model: $\cos(4\theta - 2\phi)$. (B) Separable interaction pattern with independent coefficients : $f(\theta, \phi) = \cos(4\theta) \cos(2\phi) + 0.9 \sin(4\theta) \cos(2\phi) + 0.1 \cos(4\theta) \sin(2\phi) - 0.6 \sin(4\theta) \sin(2\phi)$. (C) Geometric deformation of the interaction pattern in (2), obtained by replacing 4θ and 2ϕ with Möbius-type phase functions. (D) Field from BOUT++ tokamak plasma simulations. (E) Experimental grid-cell firing rate map.

Source: Data from [3]

In this work, we focus on oscillatory fields defined over domains that are periodic in both dimensions, allowing representation on the torus, $\mathbb{S}^1 \times \mathbb{S}^1$. This assumption is justified in several scientific settings. For instance, in magnetically confined fusion plasmas, physical quantities such as electrostatic potential or density are naturally defined over toroidal geometries [1]. In directional data, such as wind and wave orientations [2], periodicity also arises due to the circular nature of the underlying variables. Similarly, toroidal topology has been empirically observed in latent representations of grid cell activity in neuroscience [3]. In Fig. 1 we show representative examples of spatial fields illustrating a progression from idealized Fourier structures to increasingly complex and realistic spatial patterns, emphasizing the role of anisotropy. Panel (A) shows a simple anisotropic structure with a single preferred direction generated by a linear complex phase. Panel (B) allows more flexible anisotropic patterns through independent interaction coefficients. In panel (C), the same interaction coefficients are retained but the phases are replaced by Möbius-type functions, producing spatially varying anisotropy. Panels (D) and (E) show examples of spatial fields from tokamak simulations and grid-cell activity, illustrating the richer spatial organization observed in real systems.

Classical two-dimensional decompositions, such as Fourier and wavelet transforms, provide general-purpose representations but often lack directional specificity and require many components to capture localized, multicomponent, or anisotropic structures. Similarly, adaptive methods like Empirical Mode Decomposition offer greater flexibility, but their outputs can be difficult to analyze and may fail to preserve structure [4–8]. These limitations highlight key statistical challenges: modeling directionally structured oscillations, maintaining interpretability, and ensuring robustness to noise [9,10].

To address these issues, we introduce the Spatial-Frequency Modulated Möbius (S-FMM) model, a parametric framework for representing two-dimensional oscillatory fields over angular coordinates, through a finite superposition of structured oscillatory components. The model extends the Frequency Modulated Möbius (FMM) framework, originally developed for one-dimensional signals, to the spatial domain [11,12], using modulated trigonometric components with Möbius-type phase functions.

Rather than constructing the field as a simple tensor product of one-dimensional components, the model allows interactions between phase modes in each angular direction with independent coefficients. This leads to four interaction patterns (cos-cos, sin-cos, cos-sin, sin-sin), parameterized through independent coefficients. As a result, the representation captures genuinely two-dimensional modal structures that cannot arise from separable products of one-dimensional FMM signals. In signal-processing terms, complex phase representations provide only one quadrature degree of freedom, whereas two-dimensional signals may contain independent directional phase variations carrying geometric information [13]. The contrast between these two situations is illustrated in Fig. 1, where a classical Fourier interaction pattern (panel A) is compared with a representation allowing independent interaction coefficients (panel B).

The S-FMM framework incorporates several key components to meet these challenges. First, it uses a circular representation of each coordinate via complex numbers, $z = e^{i\theta}$, to naturally encode periodicity. Second, it employs a quaternionic algebraic structure with distinct imaginary units for each spatial axis, allowing the model to operate in a space consistent with toroidal topology. Third, it introduces terms that capture directional interactions, describing how variations along one axis affect oscillations along the other, which standard separable models fail to represent. Fourth, the model parameters such as amplitude, frequency, and directional phase provide an explicit description of oscillatory behavior. Finally, we propose an inference algorithm capable of recovering coherent directional modes under noise. The method is robust in practice and supports identifiability under mild constraints, providing consistent representation across structured scenarios.

While the S-FMM model is broadly applicable to structured oscillatory fields over periodic domains, in this paper we motivate and develop it primarily in the context of magnetically confined plasmas. These are high-temperature ionized gases, typically studied in the context of nuclear fusion research [14,15]. In this setting, the relevant fields are defined over toroidal geometries, with periodicity in both poloidal and toroidal directions [16,17]. Such fields often exhibit complex oscillatory patterns driven by instabilities that generate multiple coherent spatial modes [14,15]. Diagnostic measurements typically capture a superposition of such structures in the form of time series recorded across angular positions. A key challenge is to extract low-dimensional and meaningful summaries of dominant modes [18].

Standard decomposition techniques face important limitations in this context. Fourier-based approaches struggle to capture localized variations or direction-dependent interactions without resorting to many components [5,19]. Similarly, SVD and empirical

decompositions may fail to reflect the underlying physics of oscillatory fields. While machine learning methods offer flexibility, they often disregard domain-specific constraints that are essential for modeling plasma behavior [15,20,21]. These limitations highlight the value of structured models like S-FMM. Rather than introducing an incremental modification of existing spectral methods, the S-FMM framework combines ideas from nonlinear phase modeling, directional representations, and toroidal geometry into a unified approach for structured oscillatory fields.

We demonstrate the application of S-FMM to plasma dynamics using simulation data generated with BOUT++, a physics-based fluid turbulence code designed for toroidal geometries [17,22]. The potential of the S-FMM model extends beyond plasma physics. A particularly compelling application arises in neuroscience, where characterizing the structure of grid cell activity has become a central topic in systems neuroscience [3]. Additional analyses are presented in the Supplementary Material.

S-FMM should therefore be understood as a structured, low-dimensional parametric model designed to capture oscillatory patterns with interpretable geometry, rather than as a universal decomposition or a complete basis expansion on T^2 . The model is primarily intended for structured fields dominated by a small number of coherent oscillatory patterns. In particular, it is expected to be most beneficial in settings involving localized or anisotropic oscillations, directional modes with non-sinusoidal waveforms, or signals with a small number of dominant structures. Conversely, in situations where the field is well described by sinusoidal components the advantages of S-FMM are expected to be less pronounced. Within this perspective, the model provides compact and interpretable representations that can serve as informative features in data-driven analyses, including machine learning settings where geometric consistency is important.

Although S-FMM is primarily intended as a low-dimensional structured model capturing dominant oscillatory patterns, the number of modes can be increased when a more accurate reconstruction is required. In this sense, the representation may also act as a filtering or compression mechanism, allowing structured signals to be approximated using a relatively small number of parameters.

The remainder of the paper is organized as follows. Section 2 introduces the S-FMM model within a signal-plus-noise framework for representing two-dimensional oscillatory fields. Section 2.3 details the estimation algorithm, including identifiability constraints and implementation considerations. Section 3 presents empirical validation using synthetic datasets, including simulations from the BOUT++ plasma modeling framework and noisy S-FMM-generated signals. Finally, Section 4 concludes with a discussion of the results, limitations, and future research directions.

2. Methodology

2.1. Background

2.1.1. Signal complex representation for angular variables, phases and modes

The use of complex exponentials to represent oscillatory signals is standard in signal processing. For example, a real oscillation

$$x(t) = A \cos(\omega t + \phi)$$

can be written as

$$x(t) = \Re(Ae^{i(\omega t + \phi)}).$$

In this representation, the complex exponential encodes two orthogonal quadrature components, corresponding to the cosine and sine parts of the oscillation. Geometrically, the complex number represents a rotation in the complex plane, and the phase ϕ specifies the position of the signal on this circle. Thus, a one-dimensional periodic signal is naturally associated with a single phase variable and a single complex plane.

When a signal depends on two independent angular coordinates, such as $f(\theta, \phi)$, the situation changes. In many plasma physics contexts, θ and ϕ correspond approximately to the poloidal and toroidal angles, and coherent structures are described in terms of their periodicity along both directions. Classical representations of helicoidal perturbations use expressions of the form

$$e^{i(K\theta - L\phi)},$$

or equivalently $\cos(K\theta - L\phi)$, where (K, L) are the mode numbers.

From a signal-processing viewpoint, this representation combines both angular contributions into the single phase variable

$$\Phi(\theta, \phi) = K\theta - L\phi.$$

Although convenient, this compresses the phase evolution along the two directions into a single complex plane. In contrast, the underlying geometry involves two independent periodic coordinates, each associated with a circular phase space S^1 . Their joint phase space is therefore $S^1 \times S^1$, which is geometrically a torus. This suggests using a representation that preserves the phase contributions associated with the two angular directions separately, rather than projecting them onto a single phase variable.

2.1.2. Quaternionic representation

A natural way to preserve two independent phase rotations is to extend the complex algebra. Quaternions provide such an extension by introducing three imaginary units $\mathbf{i}, \mathbf{j}, \mathbf{k}$, satisfying

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = \mathbf{ijk} = -1,$$

together with the multiplication rules

$$\mathbf{ij} = \mathbf{k}, \quad \mathbf{jk} = \mathbf{i}, \quad \mathbf{ki} = \mathbf{j}, \quad \mathbf{ji} = -\mathbf{k}, \quad \mathbf{kj} = -\mathbf{i}, \quad \mathbf{ik} = -\mathbf{j}.$$

A quaternion has the form

$$q = a + b\mathbf{i} + c\mathbf{j} + d\mathbf{k}, \quad a, b, c, d \in \mathbb{R}.$$

If the phase evolution along θ and ϕ is represented associate a distinct complex plane with each direction and write $e^{\mathbf{i}K\theta}$ and $e^{-\mathbf{j}L\phi}$. This leads to the representation

$$S(\theta, \phi) = e^{\mathbf{i}K\theta} e^{-\mathbf{j}L\phi},$$

where $\mathbf{i}$ corresponds to the phase associated with the poloidal direction and $\mathbf{j}$ to that associated with the toroidal direction. Expanding the product yields

$$S(\theta, \phi) = \cos(K\theta) \cos(L\phi) + \mathbf{i} \sin(K\theta) \cos(L\phi) - \mathbf{j} \cos(K\theta) \sin(L\phi) - \mathbf{k} \sin(K\theta) \sin(L\phi).$$

This expansion shows that two independent phase rotations generate four interaction terms: one real component and three hypercomplex components. The real part recovers the classical interaction term, while the remaining terms encode quadrature interactions between the two angular directions. In this sense, quaternions arise naturally when the phase contributions associated with two periodic directions are preserved separately within a single algebraic representation.

Hypercomplex formulations have been explored in multidimensional signal processing to represent directional phase information in two-dimensional signals [13,23–25].

2.2. The S-FMM signals

2.2.1. Quaternionic S-FMM signal

Let us assume that (θ, ϕ) are angular coordinates in $[0, 2\pi)$. The quaternionic S-FMM signal is defined as a finite sum of components:

$$q(\theta, \phi) = \sum_{k=0}^K \sum_{l=0}^L Q_{kl} \exp(\mathbf{i} \Phi_k^P(\theta)) \exp(\mathbf{j} \Phi_l^T(\phi)), \quad (1)$$

where each coefficient Q_{kl} is a quaternion of the form

$$Q_{kl} = A_{kl} + \mathbf{i}B_{kl} + \mathbf{j}C_{kl} + \mathbf{k}D_{kl}, \quad A_{kl}, B_{kl}, C_{kl}, D_{kl} \in \mathbb{R}. \quad (2)$$

The phase functions $\Phi_k^P(\theta)$ and $\Phi_l^T(\phi)$ are angular modulations defined through Möbius-type transformations:

$$\Phi(\tau; \alpha, \omega) = 2 \arctan\left(\omega \tan\left(\frac{\tau - \alpha}{2}\right)\right),$$

where $\tau \in [0, 2\pi)$ is an angular variable, and $\alpha \in [0, 2\pi)$ and $\omega \in [0, 1]$ are parameters.

Although the basis functions are separable in the angular variables, the full S-FMM model allows general cross-directional coupling through the coefficient matrix Q_{kl} . In particular, each directional mode in θ can interact with any mode in ϕ , providing the representation of genuinely two-dimensional interaction patterns.

The real-valued S-FMM signal is obtained by taking the real part of the quaternionic representation:

$$f(\theta, \phi) = \Re(q(\theta, \phi)). \quad (3)$$

As a convention, we define

$$\Phi_0^P(\theta) := 0, \quad \Phi_0^T(\phi) := 0,$$

which implies

$$\cos(\Phi_0^P(\theta)) = 1, \quad \sin(\Phi_0^P(\theta)) = 0, \quad \cos(\Phi_0^T(\phi)) = 1, \quad \sin(\Phi_0^T(\phi)) = 0.$$

The Möbius modulation parameters control the geometry of the oscillations. When $\omega = 1$, the resulting waveform is sinusoidal. As ω decreases, the oscillations become increasingly peaked, providing the representation of sharp and localized angular features. These parameters are referred to as non-linear parameters, as they control the phase geometry rather than linear amplitude or frequency effects [11].

The numbers of modes in the P- and T-directions are K and L , respectively. The total number of parameters in the S-FMM model is $4KL + 4K + 4L + 1$, accounting for the quaternionic coefficients and the sets of non-linear parameters $\{\alpha_k^P, \omega_k^P\}_{k=0}^K$ and $\{\alpha_l^T, \omega_l^T\}_{l=0}^L$.

This interaction structure provides richer two-dimensional patterns than separable constructions, while preserving independent phase modulations along each angular coordinate and thereby encoding genuinely two-dimensional interactions.

This formulation therefore provides greater flexibility than classical expressions such as $\cos(K\theta - L\phi)$, allowing independent control of oscillatory behavior along each direction, supporting overlapping and anisotropic spatial patterns, and providing natural extensions to spatiotemporal signals.

2.2.2. Separable and Fourier representations

For comparison purposes it is useful to consider a separable construction obtained by removing the cross-indexed coupling structure of the quaternionic S-FMM signal. In this case the angular structure along each coordinate is modeled independently and no interaction between the two directions is introduced.

In the quaternionic framework this corresponds to representing the signal as the product of two independent one-dimensional components,

$$q_{\text{sep}}(\theta, \phi) = q_P(\theta) q_T(\phi),$$

where

$$q_P(\theta) = \sum_k p_k e^{i\Phi_k^P(\theta)}, \quad q_T(\phi) = \sum_l t_l e^{i\Phi_l^T(\phi)}.$$

This structure implies that the interaction coefficients factorize as

$$Q_{kl} = p_k t_l,$$

so that the matrix of interaction coefficients has rank one. Consequently, the coupling between oscillatory modes along the two angular directions is restricted.

A different type of constraint arises in classical Fourier representations. A Fourier expansion on the torus can be written in complex form as

$$f(\theta, \phi) = \sum_{k,l} c_{kl} e^{i(k\theta + l\phi)}.$$

In many physical applications, such as plasma physics, oscillatory structures are often described through helicoidal modes associated with a particular pair of integers (K, L) , written as

$$\Re(c_{kl} e^{i(K\theta - L\phi)}) = A \cos(K\theta - L\phi) + B \sin(K\theta - L\phi),$$

which correspond to a single linear phase combination $K\theta - L\phi$. When expanded in trigonometric form, these modes generate the interaction patterns

$$\cos(K\theta) \cos(L\phi), \quad \sin(K\theta) \cos(L\phi), \quad \cos(K\theta) \sin(L\phi), \quad \sin(K\theta) \sin(L\phi),$$

but with coefficients linked by linear relations, since they originate from the same complex coefficient.

More generally, the Fourier series can be written directly in real trigonometric form as

$$f(\theta, \phi) = \sum_{k,l} \left[A_{kl} \cos(k\theta) \cos(l\phi) + B_{kl} \sin(k\theta) \cos(l\phi) + C_{kl} \cos(k\theta) \sin(l\phi) + D_{kl} \sin(k\theta) \sin(l\phi) \right],$$

where the four interaction coefficients are independent. While both formulations describe the same class of periodic functions (see, e.g., [5]), the helicoidal representation emphasizes a single linear phase $K\theta - L\phi$, whereas the real trigonometric formulation makes explicit the four interaction channels between the two angular directions.

The constructions discussed above reveal two structural limitations. Separable models impose a rank-one interaction structure, restricting the coupling between modes along the two angular directions. Helicoidal modes impose linear relations among the four trigonometric interaction coefficients, whereas the full Fourier representation allows them to vary independently.

The S-FMM representation extends both classical Fourier constructions and separable models for signals defined on two angular coordinates. In this framework the oscillatory interactions along the two directions are described through the quaternionic coefficients described in Eq. (2), which naturally encode the four interaction components.

In addition, the linear sinusoidal phases are replaced by Möbius-type phase functions, providing a flexible description of non-sinusoidal phase evolution. More explicitly, in the S-FMM formulation, when $\omega_k^P = \omega_l^T = 1$ for all modes, the phase functions reduce to linear angular phases and the model recovers the standard real Fourier basis on the torus.

The quaternionic formulation makes this relationship particularly transparent. In the sinusoidal limit the phase factors $\exp(i\Phi_k^P(\theta))$ and $\exp(j\Phi_l^T(\phi))$ generate the usual harmonic oscillations along each angular coordinate, while the quaternionic coefficients determine the amplitudes of the four real interaction patterns.

These interaction structures are illustrated in Fig. 1. Panel (A) shows a single helicoidal interaction. Panel (B) illustrates a representation in which the four interaction components can vary independently while the phases remain linear. Panel (C) uses the same interaction coefficients as panel (B), but replaces the linear phases by Möbius-type phase functions in the two angular directions.

2.2.3. S-FMM alternative formulations

Using standard trigonometric identities, we have

$$\Re[Q \exp(i\Phi^P(\theta)) \exp(j\Phi^T(\phi))] = A \cos(\Phi^P(\theta)) \cos(\Phi^T(\phi)) + B \sin(\Phi^P(\theta)) \cos(\Phi^T(\phi)) \\ + C \cos(\Phi^P(\theta)) \sin(\Phi^T(\phi)) + D \sin(\Phi^P(\theta)) \sin(\Phi^T(\phi)),$$

where $Q = A + iB + jC + kD$, with $A, B, C, D \in \mathbb{R}$.

This expression applies to each S-FMM component and clarifies how the model combines cosine and sine terms along both angular directions. For brevity, we henceforth refer to the S-FMM signal as the real part of its quaternionic representation.

The following proposition provides an equivalent formulation of the S-FMM signal that motivates the estimation strategy. The proof is deferred to the Supplementary Material.

Proposition. Let $f(\theta, \phi)$ be an S-FMM signal defined on $[0, 2\pi)^2$. Let $\{\Phi_k^P(\theta)\}_{k=0}^K$ and $\{\Phi_l^T(\phi)\}_{l=0}^L$ be fixed phase functions in the P- and T-directions. Then the following statements are equivalent:

- (a) $f(\theta, \phi) = \sum_{k=0}^K \sum_{l=0}^L [A_{kl} \cos(\Phi_k^P(\theta)) \cos(\Phi_l^T(\phi)) + B_{kl} \sin(\Phi_k^P(\theta)) \cos(\Phi_l^T(\phi)) \\ + C_{kl} \cos(\Phi_k^P(\theta)) \sin(\Phi_l^T(\phi)) + D_{kl} \sin(\Phi_k^P(\theta)) \sin(\Phi_l^T(\phi))]$.
- (b) $f(\theta, \phi) = M_\phi + \sum_{k=1}^K R_k(\phi) \cos(\Phi_k^P(\theta) + \beta_k(\phi))$,
where the collection: $\{M_\phi, R_k(\phi) \cos(\beta_k(\phi)), R_k(\phi) \sin(\beta_k(\phi)) \text{ for } k = 1, \dots, K\}$
is modeled as a multichannel FMM series in ϕ , expressed as linear combinations of $\cos(\Phi_l^T(\phi))$ and $\sin(\Phi_l^T(\phi))$, for $l = 1, \dots, L$.

This formulation reveals a double multichannel structure. For fixed ϕ , expression (b) defines a univariate FMM in θ . Evaluating $f(\theta, \phi)$ across multiple values of ϕ yields a collection of such signals, which together form a multichannel FMM along the P-direction. The corresponding quantities in ϕ are likewise modeled as a multichannel FMM in the T-direction.

2.3. Estimation strategy for S-FMM models

Let $X(\theta_i, \phi_j)$, for $i = 1, \dots, n$ and $j = 1, \dots, m$, denote the observed data over a grid of angular coordinates. We assume a signal-plus-noise model of the form

$$X(\theta_i, \phi_j) = f(\theta_i, \phi_j) + \varepsilon_{ij},$$

where $f(\theta, \phi)$ is the underlying S-FMM signal and $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$ are independent, identically distributed Gaussian errors. Under this assumption, parameter estimation can be formulated as a least-squares problem. Although the independent Gaussian noise model is adopted for simplicity, the proposed estimation procedure is robust to more general noise structures. In particular, its performance under spatially correlated noise, which is more representative of real-world scenarios, is assessed in the simulation study presented in Section 3.

To ensure parameter identifiability and avoid degenerate solutions due to label switching, we impose ordering constraints on the phase location parameters, $\alpha_1^P \leq \alpha_2^P \leq \dots \leq \alpha_K^P$ and $\alpha_1^T \leq \alpha_2^T \leq \dots \leq \alpha_L^T$. As a consequence, the shape parameters ω_k^P and ω_l^T are uniquely associated with their corresponding directional modes, yielding a well-posed estimation problem.

The double multichannel structure established in Section 2.2.3 motivates a two-stage estimation approach. One angular direction is treated first, holding the other fixed and applying standard multichannel FMM techniques. The recovered components are then expanded along the second direction using a subsequent multichannel fit. The roles of θ and ϕ can be interchanged depending on the application; for computational efficiency, it is often preferable to begin with the direction requiring fewer modes.

More concretely, we propose a five-step estimation procedure that includes a data-driven selection of the number of oscillatory modes. This strategy balances computational cost and estimation accuracy by separating the nonlinear and linear blocks of the model.

2.3.1. Sequential parameter estimation

1. Mode number selection

Estimating the number of dominant modes along each angular direction depends on the goals of the analysis, such as compact representation or high reconstruction accuracy. In plasma data analysis, this step is often addressed heuristically using polar plots, phase-crossing counts, or inspection of the leading singular vectors from singular value decomposition (SVD) [18]. While such methods provide qualitative guidance, they may yield ambiguous results. In our approach, the leading left and right singular vectors, corresponding to the P- and T-directions, are modeled using one-dimensional FMM representations. The goodness-of-fit (R^2) of these models provides a quantitative criterion for selecting the numbers of modes K and L . A standard strategy is to increment K or L until the marginal improvement in fit becomes negligible.

2. Estimation of nonlinear parameters along the first direction

The parameters governing oscillations along the P-direction are first estimated as a multichannel FMM model [12,26]. Then, we consider the columns of the data matrix, corresponding to different ϕ values, as individual channels.

3. Estimation of nonlinear parameters along the second direction.

Given the estimates obtained for the first direction, the parameters governing oscillations along the second direction are subsequently estimated. This step exploits Section 2.2.3, which shows that, in the case of taking T as the first direction, the decomposition reduces to a multichannel FMM representation of the functions $\{M_\phi, R_k(\phi) \cos(\beta_k(\phi)), R_k(\phi) \sin(\beta_k(\phi))\}_{k=1}^K$.

4. Estimation of linear parameters.

Conditioned on the nonlinear parameters, the remaining linear parameters are estimated using standard linear least-squares techniques.

5. Joint refinement.

A final joint optimization step refines all parameters, using the previous estimates as initialization.

If the final model does not achieve satisfactory reconstruction quality, the numbers of modes K and L may be revised and the estimation procedure repeated. A detailed description of the full estimation workflow, including computation times for the stages and illustrative examples, is provided in the Supplementary Material.

2.3.1.1. Identifiability considerations. In classical Fourier expansions the phases are fixed by the harmonic frequencies and are therefore not parameters of the model. The only estimated quantities are the linear Fourier coefficients. In contrast, the S-FMM formulation introduces explicit phase parameters through the functions $\Phi_k^P(\theta)$ and $\Phi_l^T(\phi)$, which are controlled by the parameters (α, ω) . These parameters determine the location and shape of the oscillatory structures.

The parameters governing the phase geometry, in particular the phase locations α and shape parameters ω , determine the location and sharpness of the dominant oscillatory structures. The ordering constraints imposed on the location parameters α_k^P and α_l^T are included to prevent label switching between modes, ensuring that each component corresponds to a well-defined region in the angular domain. Simulation experiments reported in Section 3 show that these parameters are estimated in a stable and reproducible manner under repeated estimations and varying noise levels. This supports the practical interpretability of the geometric parameters of the model.

3. Numerical experiments

We conduct a series of numerical experiments to assess the empirical performance of the S-FMM model under controlled and realistic conditions.

First, synthetic data are generated to evaluate practical identifiability and robustness of the estimation procedure under noise. Different noise types and levels are considered to assess stability under both idealized and physically plausible perturbations.

Second, the performance of S-FMM is benchmarked against two standard baselines, namely two-dimensional Fourier analysis and SVD-based decomposition. This comparison is carried out using three representative synthetic datasets generated with the BOUT++ plasma simulation framework: the Hasegawa–Wakatani (HW) model, a synthetic blob model, and the ELM-pb (Edge Localized Modes, peeling–ballooning type) model [17].

Additional simulation scenarios are presented in the Supplementary Material, providing further validation of the proposed framework. In a separate applied example, also reported in the Supplementary Material, we analyze the spatial firing rate map of a grid cell (see Fig. 1). This example illustrates the applicability of the S-FMM framework beyond plasma physics, in a setting involving latent toroidal topology.

3.1. Synthetic noisy data

Synthetic data are generated using S-FMM parameters estimated from the ELM-pb dataset, which is analyzed in detail in the next subsection. Specifically, we consider parameters corresponding to $t = 30$, a time point exhibiting high reconstruction accuracy and representative spatial complexity. The model's ability to recover these parameters is then evaluated under varying levels of additive noise. Fig. 3(c) displays the original data, the SVD and two-dimensional Fourier reconstructions, and the S-FMM reconstruction used as the basis for generating the synthetic signals.

Two types of noise are added to the clean simulated signals: (i) white Gaussian noise, representing unstructured measurement errors, and (ii) spatially correlated noise, obtained by applying a Gaussian blur to white noise, mimicking perturbations with local spatial coherence commonly observed in experimental diagnostics. For each noise type, two levels are considered, referred to as low and moderate, both of which visibly affect the spatial patterns, as illustrated in Fig. 2.

This setup provides a systematic evaluation of the estimation procedure by comparing the original model parameters with those recovered from noisy synthetic signals. Mean squared errors (MSE) are computed for the ω parameters, wrapped MSE (wMSE) for the angular location parameters α to account for circular topology [27], and explained variance (R^2). Each scenario is repeated over 100 trials. The results are summarized in Table 1.

As reported in Table 2, the MSE values remain consistently low for both α and ω parameters, with only moderate degradation observed as noise levels increase. The corresponding average R^2 values, ranging from approximately 0.98 under low noise to 0.89 under moderate noise, indicate that the S-FMM model achieves high-fidelity reconstructions even under challenging conditions.

Additional results under structured noise and increased signal complexity are reported in the Supplementary Material, including scenarios with $K = L = 2$ and $K = L = 4$. These experiments involve more complex oscillatory patterns, yet lead to conclusions consistent with those reported here.

Table 1

True values, confidence intervals (CI), and MSEs for selected parameters under structured and white noise at low and moderate levels. Subscripts (1), (2) denote distinct components where applicable. Values rounded to 4 decimals.

Parameter	True Value	CI Lower	CI Upper	MSE
White Noise — Low (Mean R^2 : 0.9871)				
$\alpha_p^{(1)}$	0.2502	0.2393	0.2643	0.0001
$\omega_p^{(1)}$	0.5558	0.5510	0.5622	0.0000
$\alpha_T^{(1)}$	6.1574	6.1334	6.1798	0.0002
$\alpha_T^{(2)}$	0.1690	0.1516	0.1867	0.0001
$\omega_T^{(1)}$	0.4091	0.3998	0.4236	0.0000
$\omega_T^{(2)}$	0.3236	0.3137	0.3297	0.0000
White Noise — Moderate (Mean R^2 : 0.8943)				
$\alpha_p^{(1)}$	0.2502	0.1966	0.2913	0.0006
$\omega_p^{(1)}$	0.5558	0.5417	0.5728	0.0001
$\alpha_T^{(1)}$	6.1574	6.0484	6.2395	0.0022
$\alpha_T^{(2)}$	0.1690	0.1186	0.2323	0.0010
$\omega_T^{(1)}$	0.4091	0.3723	0.4509	0.0004
$\omega_T^{(2)}$	0.3236	0.2929	0.3453	0.0002
Structured Noise — Low (Mean R^2 : 0.9878)				
$\alpha_p^{(1)}$	0.2502	0.2122	0.2944	0.0006
$\omega_p^{(1)}$	0.5558	0.5409	0.5761	0.0001
$\alpha_T^{(1)}$	6.1574	6.0903	6.2279	0.0015
$\alpha_T^{(2)}$	0.1690	0.1281	0.2114	0.0005
$\omega_T^{(1)}$	0.4091	0.3863	0.4483	0.0003
$\omega_T^{(2)}$	0.3236	0.2983	0.3389	0.0001
Structured Noise — Moderate (Mean R^2 : 0.8998)				
$\alpha_p^{(1)}$	0.2502	0.0765	0.3780	0.0065
$\omega_p^{(1)}$	0.5558	0.5119	0.6077	0.0009
$\alpha_T^{(1)}$	6.1574	5.8773	6.4375	0.0147
$\alpha_T^{(2)}$	0.1690	0.0605	0.3082	0.0047
$\omega_T^{(1)}$	0.4091	0.3206	0.5301	0.0032
$\omega_T^{(2)}$	0.3236	0.2561	0.3934	0.0018

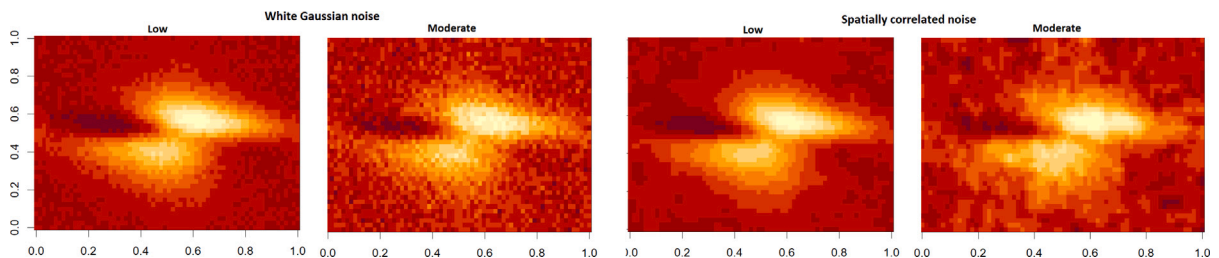


Fig. 2. ELM-pb simulations under noisy conditions. Left panels correspond to low noise levels, while right panels show moderate noise levels.

3.2. BOUT++ simulated data analysis

The plasma simulation example should be interpreted primarily as a proof-of-concept illustrating the representational capabilities of the S-FMM model on a physically motivated dataset. A detailed physical interpretation of the recovered components is beyond the scope of the present work.

All datasets were generated using the BOUT++ simulation framework under different configurations designed to reflect varying spatial structures of the perturbed plasma density field (n), a physically meaningful and diagnostically accessible quantity that

Table 2

Performance comparison of rank-1 SVD, two-dimensional Fourier, and S-FMM decompositions on BOUT++ simulations. Results are shown for different physical regimes and time instances t , reporting mesh size, number of angular modes (K, L), coefficient of determination R^2 (in %), and total number of model parameters.

Case	Method	Mesh	K	L	R^2 (%)	# Params
HW, $t = 10$	SVD	256×260	–	–	47	515
	Fourier		6	11	30	336
	S-FMM		5	10	94	261
HW, $t = 20$	SVD	256×260	–	–	67	515
	Fourier		6	11	10	336
	S-FMM		5	10	82	261
HW, $t = 30$	SVD	256×260	–	–	58	515
	Fourier		6	11	33	336
	S-FMM		5	10	84	261
Blobs, $t = 10$	SVD	57×71	–	–	86	127
	Fourier		3	3	67	49
	S-FMM		2	2	97	33
Blobs, $t = 20$	SVD	77×61	–	–	38	137
	Fourier		3	3	49	49
	S-FMM		2	2	85	33
Blobs, $t = 30$	Fourier	95×89	5	5	52	121
	S-FMM		4	4	94	97
	SVD		–	–	25	183
ELM-pb, $t = 10$	Fourier	68×42	3	3	47	49
	S-FMM		2	2	85	33
	SVD		–	–	64	109
ELM-pb, $t = 20$	Fourier	68×20	3	2	50	36
	S-FMM		2	1	95	21
	SVD		–	–	97	87
ELM-pb, $t = 30$	Fourier	68×42	3	2	68	36
	S-FMM		2	1	99	21
	SVD		–	–	77	109
ELM-pb, $t = 30$	Fourier	68×42	3	2	48	36
	S-FMM		2	1	97	21
	SVD		–	–	64	109

captures key features of turbulence and coherent mode activity. The two spatial directions in the simulation domain are treated as angular and periodic, although they do not necessarily correspond exactly to the physical toroidal and poloidal angles.

We focus on fixed-time slices of the plasma density field, represented as two-dimensional angular maps defined over both spatial coordinates. To enhance the robustness of the modal decomposition, we use temporally averaged snapshots of the data centered at each target time point. Specifically, we average over a time window from $t - 3$ to $t + 3$, which smooths out incoherent fluctuations while preserving the dominant spatial structures. Furthermore, we restrict the analysis to a subdomain exhibiting significant spatial variation. This region is identified using the leading singular vectors from the SVD, which concentrate the signal energy and reduce boundary effects.

We apply this analysis framework to three benchmark cases with increasing spatial complexity:

- **HW (Hasegawa–Wakatani):** a simple model that produces smooth and regular wave patterns.
- **Blobs:** synthetic, localized patterns that appear intermittently and are highly fragmented in space.
- **ELM-pb:** a more complex simulation with structured patterns that vary across space and time.

The selection of HW, Blobs, and ELM-pb provides a balanced framework for evaluating modeling techniques across regimes ranging from smooth turbulence to structured and intermittent dynamics. Our objective is to explore the spatial organization of the signal and to assess the ability of different decomposition methods — namely two-dimensional Fourier analysis, SVD, and S-FMM — to represent this structure and capture its essential features.

For comparative purposes, the two-dimensional Fourier model includes one additional component in each angular direction in order to ensure a fair comparison in terms of model complexity. The SVD baseline is included as a minimal linear reference model, and we therefore consider a rank-1 reconstruction to maintain comparable complexity with the proposed S-FMM representation and the other compact baselines considered in this study. Notably, the SVD consistently reveals a single dominant mode across all averaged fields, supporting the low-rank structure assumption underlying this comparative setup.

Our goal is not to benchmark the proposed representation against all existing signal decomposition frameworks. Instead, we focus on compact harmonic baselines that provide natural reference models for periodic fields and allow meaningful comparisons in terms of parsimonious representations.

3.2.1. Results

We compare the performance of the three decomposition methods, SVD, two-dimensional Fourier analysis, and S-FMM, using both quantitative metrics and visual inspection across all simulation scenarios. The coefficient of determination (R^2) is used as the main metric, since the focus of the experiments is on reconstruction accuracy under parsimonious model representations. Table 2 summarizes the R^2 values, the model dimensions (K , L), and the number of parameters used by each method at representative time frames. The SVD approximations use only the first singular component (rank-1). Overall, the results indicate that the S-FMM model consistently outperforms both alternatives, often achieving higher reconstruction accuracy with substantially fewer parameters.

Hasegawa–Wakatani (HW). This dataset features wave-like, periodic structures, making it well suited for evaluating each method's ability to represent coherent directional oscillations. At early times (e.g., $t = 10$), the field appears visually disordered due to the superposition of unstable modes, as shown in Fig. 3(a). The SVD reconstruction captures a broad trend but lacks directional sharpness, while the two-dimensional Fourier approximation introduces interference artifacts. In contrast, S-FMM closely approximates the shape and orientation of the dominant wave patterns, preserving fine details with lower model complexity. Quantitatively, S-FMM achieves $R^2 = 94\%$, compared to 47% for SVD and 30% for the Fourier method.

Blobs. This case involves highly localized and asymmetric structures that are difficult to capture using global bases. As illustrated in Fig. 3(b) at $t = 20$, the SVD approximation oversmooths the field and fails to isolate key peaks ($R^2 = 38\%$), while the Fourier representation introduces spurious oscillations and misalignment ($R^2 = 49\%$). S-FMM captures the main localized features with greater fidelity and compactness. Increasing the model complexity from $K = L = 2$ to $K = L = 4$ allows sharper structures to be recovered, improving the reconstruction accuracy from $R^2 = 85\%$ to $R^2 = 94\%$.

ELM-pb. In this more complex scenario (Fig. 3(c)), the field at $t = 30$ exhibits structured and nontrivial modulation. The SVD reconstruction reflects the overall shape but fails to recover important spatial details ($R^2 = 77\%$), while the Fourier approximation introduces noticeable distortion ($R^2 = 48\%$). S-FMM reconstructs both the global structure and fine-scale modulation with high fidelity, achieving $R^2 = 97\%$ using only 25 parameters.

Across all scenarios, S-FMM demonstrates a favorable trade-off between reconstruction accuracy and model compactness, clearly outperforming SVD and two-dimensional Fourier methods. While SVD minimizes reconstruction error in a linear sense, its components lack directional specificity and physical interpretability. Fourier methods rely on rigid bases and often require many components to achieve comparable accuracy. In contrast, S-FMM achieves high-fidelity reconstructions with fewer components while preserving directional structure and physical relevance.

The results obtained from BOUT++ simulations highlight the ability of S-FMM to extract coherent and interpretable spatial structure from complex plasma fields. Fully exploiting the physical meaning of the recovered modes and parameters will require close collaboration with domain experts in plasma physics, which we identify as an important direction for future work.

4. Discussion

This work introduces the S-FMM model, a flexible framework for representing structured two-dimensional oscillatory fields on periodic domains. By combining Möbius-based phase functions with a quaternionic representation, S-FMM captures directional dependencies and complex phase relationships across spatial dimensions, offering clear advantages over classical spectral methods. Although only the real part of the quaternionic signal is observed, the imaginary components provide a structured encoding of directional variation along the two angular coordinates, providing a richer and more interpretable description of spatial organization.

Furthermore, we propose a five-stage estimation procedure that includes a data-driven strategy for selecting the number of modes in each direction. Numerical experiments across a range of noise levels and spatial configurations demonstrate robust empirical behavior. Validation on representative BOUT++ plasma simulations, spanning smooth turbulent fields, localized blob-like structures, and nonlinear edge dynamics, shows that S-FMM consistently achieves high reconstruction accuracy with a compact set of parameters. In all considered scenarios, S-FMM outperforms two-dimensional Fourier and SVD-based decompositions, both in terms of reconstruction quality and structural interpretability.

By capturing dominant spatial modes using a compact and structured basis, S-FMM is well suited for diagnostic and exploratory analysis tasks, including noise filtering, dimensionality reduction, and structurally signal reconstruction. The resulting parsimonious representations are also attractive for data-driven modeling pipelines, where preserving structure and interpretability is essential. In this context, an interesting direction for future research is the integration of S-FMM concepts into learning-based approaches. For example, inspired by recent developments in neural operator models for plasma dynamics [28], one could envision architectures that learn the evolution of S-FMM modal coefficients directly from data, combining the transparency of the proposed framework with the flexibility of deep learning.

A current limitation of the approach is the computational cost of the estimation procedure, which may become significant for very high-resolution spatial grids or long temporal sequences. This motivates future work on accelerating the optimization, for instance through parallel implementations or more efficient initialization strategies.

Although the present study focuses on fixed-time spatial fields, the S-FMM framework naturally extends to dynamic settings. Applying the estimation procedure over time yields trajectories of model parameters that characterize the evolution of dominant oscillatory modes and may provide insight into regime transitions or correlations with external physical variables.

Finally, although this work is motivated by problems in plasma physics, the applicability of S-FMM extends well beyond this domain. As illustrated by the grid cell example, the model captures structured periodic patterns in neural systems, where latent toroidal geometry has been observed [3]. More broadly, S-FMM is relevant to a wide range of problems involving spatially organized oscillations on periodic or angular domains, including geophysics, fluid dynamics, and climate modeling. In these settings, the quaternionic formulation offers a compact and flexible representation for signals defined on circular coordinates or curved surfaces, providing a unifying perspective for modeling structured spatial, and potentially spatiotemporal, phenomena.

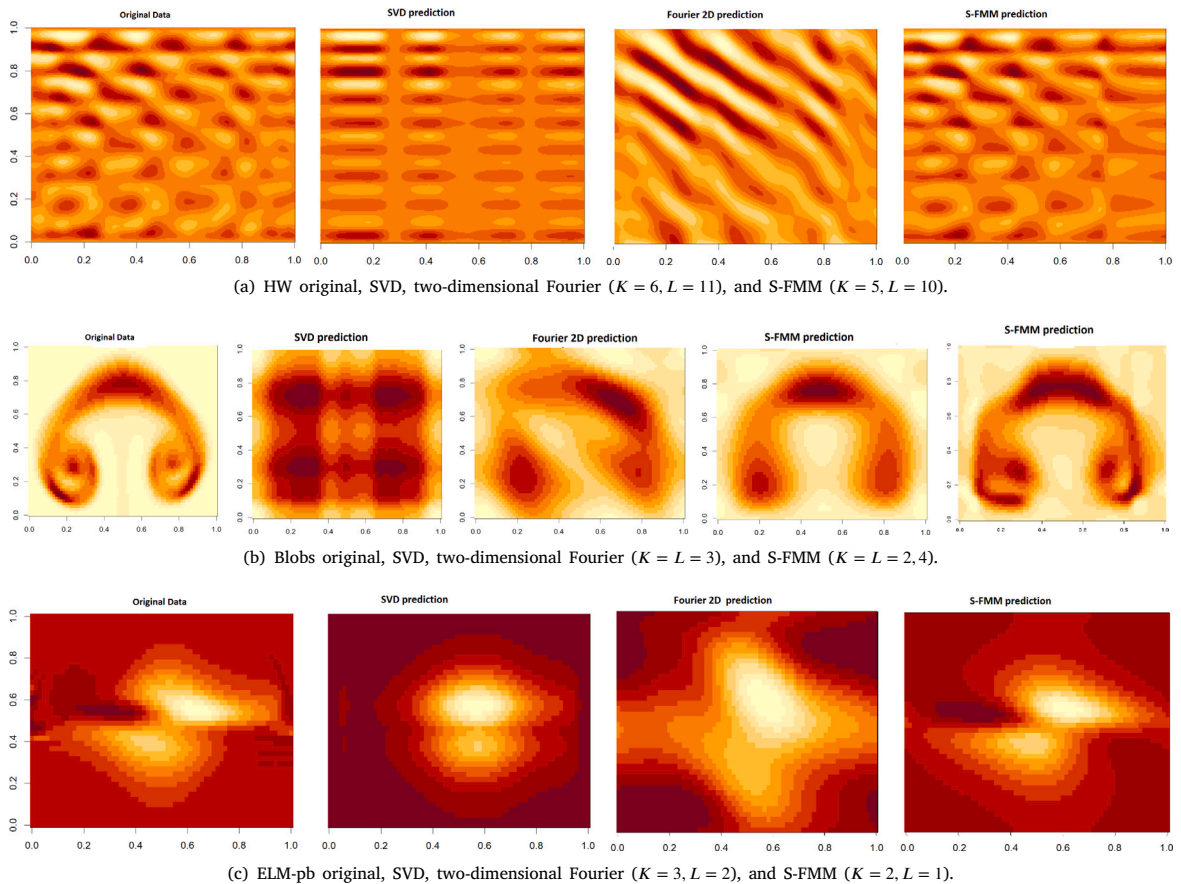


Fig. 3. Model predictions across BOUT++ simulations using SVD, two-dimensional Fourier analysis, and S-FMM. (a) HW ($t = 10$), (b) Blobs ($t = 20$), and (c) ELM-pb ($t = 30$).

CRediT authorship contribution statement

Christian Canedo: Writing – review & editing, Supervision, Software. **Cristina Rueda:** Writing – original draft, Software, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary Material: A PDF with Supplementary Material.

Python and R Code: <https://github.com/FMMGroupVa>. Containing the Python and R codes used to generate the S-FMM predictions and run the simulations described in this paper. It is based on the module presented in [26].

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.chaos.2026.118532>.

Data availability

The dataset analyzed originates from [3], and the code used to compute the firing rate maps from this data is also provided in the repository. Plasma simulations were performed using the BOUT++ framework [17].

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